1 Application software is run on thin client computers. A thin client computer is a very low-powered computer connected to a powerful central server. The operating system and all the applications run within a virtual machine on the server. The thin client computer will only display the output of the virtual machine and capture and send input to the virtual machine.

State **one** advantage of running the application software within the virtual machine.

---

[1]

2 Two roles of an operating system are to handle interrupts and manage scheduling.

i. Describe **two** other roles of an operating system.

1 ____

____

____

____

2 ____

____

____

____ **[4]**

ii. Draw a line to match each scheduling algorithm to the correct description.

Scheduling Algorithm
Round Robin
First come first served
Multi-level feedback queues
Shortest job first
Shortest remaining time

Description
Splits processes into different priority queues based on the amount of processor time they need. It allows them to move between the queues as their characteristics change
Selects the process that takes the shortest amount of time to complete. The processes are run until they are fully complete
Each process is allocated a fixed amount of CPU time. If the process is not complete it will be suspended and the next process will start
Each process is given equal priority and they are processed in the order they arrive
Selects the process that takes the shortest amount of time. The process can be suspended if another shorter process is added

[5]

3(a) A secondary school is upgrading their computers. They decide to install "thin client" computers. A thin client computer is when users access their computer in the usual way with a keyboard, mouse and monitor. However, all processing takes place on the virtual machine on a server rather than the computer at their desk.

i. Describe **one** advantage of using virtual machines in this way.

____

____

____

____ [2]

ii. Describe **one** disadvantage of using virtual machines in this way.

____

____

____

____ [2]

(b) Each virtual machine will run an operating system. One type of operating system is multi-user.

Describe **two** other types of operating system and give an example of where each may be used.

TYPE 1

Description

Example

TYPE 2

Description

Example

[6]

4(a) A small manufacturing business uses networked computers with closed source application software installed.

One computer owned by the business monitors critical-safety features of manufacturing. All input data must be processed within a predictable timescale of a fraction of a second.

i. State the type of operating system that should be used by this computer.

[1]

ii. Give the name of **three** other types of operating system, and for each state its purpose.

Type 1

Purpose 1

Type 2

Purpose 2

Type 3

Purpose 3

[6]

(b) When a device such as a keyboard or printer requires attention from the CPU, an interrupt is raised.

Explain how an operating system deals with an interrupt.

[3]

(c) Memory management is a key function of an operating system. Explain how an operating system can manage the memory available to applications and why doing so is important.

You should include the following in your answer:

- the different actions carried out by an operating system to manage memory
- how memory that is being managed can be split up
- why memory management is important.

[9]

5 A computer system will contain several input and output devices.

Explain the role of device drivers when using input and output devices on a computer system.

[2]

6 Anika's computer runs a multi-tasking operating system. She has access to a printer and a broadband internet connection through a wireless connection. The operating system uses scheduling algorithms such as first come first served and round-robin.

i. Explain why the computer's operating system uses a first come first served algorithm when sending documents to the printer.

[2]

ii. Explain why the computer's operating system uses a round-robin algorithm for allocating processor time.

[3]

iii. Describe **one** other scheduling algorithm.

[2]

7(a) One role of an operating system is the Interrupt Service Handler. This allows processes being executed by the CPU to be interrupted.

i. One example of an interrupt would be removing an external hard disk drive from a computer.

State why this would need to interrupt the current fetch-decode-execute cycle of the CPU.

____

____ [1]

ii. Interrupt Service Handlers make use of a stack data structure.

Describe how a stack is used when handling interrupts.

____

____

____

____ [2]

(b) Arnold has a router. It will receive data packets from other computers on Arnold's network or the internet and then route them on to the next step.

The scheduling algorithm Arnold's router uses is First Come First Served.

i. State the name of **one** other scheduling algorithm.

____

____ [1]

ii. Explain why First Come First Served is a suitable scheduling algorithm for Arnold's router.

____

____

____

____ [2]

(c) One role of an operating system is to manage the computer's memory.

Two types of memory management are paging and segmentation.

Describe **one** difference between paging and segmentation.

[2]

8 Julie is a university student. She is considering buying a laptop to help with her studies both at home and university. Her friend has told her she will need to choose an operating system to run on her laptop.

Two functions of an operating system are memory management and scheduling.

State **two** other functions of an operating system.

1

2

[2]

9 A company makes anti-virus software.

Anti-virus software is an example of a utility.

When running the anti-virus software, an operating system uses a scheduling algorithm to determine an allocation of CPU time to the anti-virus software.

Explain why a First Come First Served scheduling algorithm would **not** be suitable in this situation.

[2]

10 A taxi firm is investigating replacing its drivers with self-driving cars.

Explain why the self-driving system will use a real-time operating system.

-- [3]

11 Each computer in a firm with high end computers with high performance CPUs, GPUs and large amount of RAM has a multi-tasking operating system installed.

i. State the name of and describe **two** methods that the operating system can use to divide the contents of RAM.

Method 1

Name --

Description --

Method 2

Name

Description

[4]

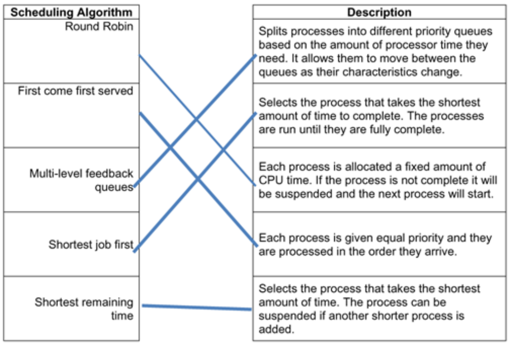
ii. Explain, giving an example, why the firm's computers use operating systems capable of multi-tasking.

[2]

END OF QUESTION PAPER

Question			Answer/Indicative content	Marks	Guidance
1			1 mark for: • Reduced hardware is needed on computers // reduced hardware costs • Improved security by keeping all software running from one physical device • Easier/cheaper to manage as only one physical device runs the programs. • Can add/remove resources/memory/processes to the VM • If it is infected by malware it can be deleted and recreated // the rest of the system isn't affected/is protected against malware • No direct access to hardware • Resources can be used more flexibly between the machines	1	<u>Examiner's Comments</u> There were some good responses to this question with many giving the response that Malware would not affect the server. Some talked about being able to try out different operating systems which was not relevant to this scenario.
			Total	1	

Question			Answer/Indicative content	Marks	Guidance
2		i	2 marks from each group to max 4: • Resource/memory management e.g. • Moving data between RAM and secondary storage/ virtual memory // paging and/or segmentation • Allocating/deallocating memory • Manage hardware/peripherals e.g. • Tracking all devices connected to the system • Device drivers • File management e.g. • Storing files in secondary storage • Searching for //copying // moving // renaming files/folders • Security/user management e.g. • Controlling who can access the system //Managing user profiles • Controlling who can access certain resources on the system // Managing access rights • Provide a user interface e.g. • Allowing the user to interact with the software/hardware/computer • Providing utilities e.g. • Used to monitor // manage // maintain the computer • To manage the security • Providing a platform on which to run software e.g. • Allows additional software to be installed on the computer • To allow the user to complete additional tasks	4	DNA handle interrupts or manage scheduling. Allow any reasonable description that matches with the role. Mark in groups. Examiner's Comments This was generally well answered and the majority of candidates could give two roles. Less successful candidates were then unable to follow through with a reasonable description. More successful candidates were able to give two roles plus relevant and correct descriptions. Some described scheduling and interrupts which were given in the question and gained no marks.

Question			Answer/Indicative content	Marks	Guidance
		ii	1 mark for each correct match. 	5	<u>Examiner's Comments</u> Generally well answered and most candidates gained full marks. Those that did not tended to confuse shortest remaining time with shortest job first but were still able to gain three marks.
			Total	9	

Question			Answer/Indicative content	Marks	Guidance
3	a	i	• Allows for more efficient use of computing resources • ... Less likely to have workstations run idle • ... can allocate more resources to power users dynamically Or • Thin clients are cheaper than desktops... • ...which could save the school money. Or • Thin clients are often physically smaller than desktops... • ...allowing more desk space/workspace for staff/students.	AO2.1 (2)	Accept other valid reasons with a suitable justification <u>Examiner's Comments</u> This question was generally not answered well. Candidates did not make sure that their responses were specific to thin clients.
		ii	• Can cause network overload... • ...if the network does not have sufficiently good bandwidth • Could create process lag... • ...If lots of large processes run at the same time • If a virtual machine hangs/crashes the user is unable to reset it... • ...whereas they can physically reboot on a local machine. • Thin-clients may not be compatible with some peripherals... • ...meaning some of the school's peripherals become unusable.	AO2.1 (2)	Mark in pairs <u>Examiner's Comments</u> As with Question 3 (a) (i), candidates did not make sure that their responses were specific to thin clients and therefore lost marks.

Question			Answer/Indicative content	Marks	Guidance
	b		• **Multi Tasking** ... • ... is optimised to deal with multiple processes simultaneously/concurrently • ...Can run multiple programs/tasks at the same time Example, e.g. • Home/personal computer • **Distributed** ... • ... Designed to run the **same** task over multiple machines Example, e.g. • SETI or other cloud-based processes// Clusters of desktops used for rendering/calculations. • **Embedded** ... • ... Will have a restricted set of operations (to conserve resources) • ... is read only Example, e.g. • Domestic appliance such as a washing machine • **Real-Time** ... • ...optimised to handle and respond to inputs with no wait/within a guaranteed response time/immediate. Example, e.g. • Used in "mission critical" systems such as aircraft control.	AO1.1 (3) AO1.2 (3)	1 mark for type of OS 1 mark for description 1 mark for suitable example Allow other suitable types of operating system. <u>**Examiner's Comments**</u> This question was generally answered well, although some candidates were too vague in their descriptions of real time and distributed operating systems in particular. Exemplar 2 **TYPE 1** Description real-time operating system, which is used when a response is required within a guaranteed time frame Example Intensive Care Units for use when patients need immediate assistance / treatment **TYPE 2** Description embedded system, which is stored in the BIOS and has limited functionality, as it is used to perform specific tasks. Example a smart washing machine, as it's instructions aren't likely to change The candidate has been given full marks for this response as they have been clear that real time operating systems respond in a guaranteed time frame, with a suitable example of their use. Some candidates were too vague in their response for descriptions of a real time operating system with answers such as 'responds quickly'.
			Total	**10**	

Question			Answer/Indicative content	Marks	Guidance
4	a	i	Real time	1	Correct answer only __Examiner's Comments__ This question was generally well answered.
		ii	• __Multi-tasking__… • …runs multiple **programs at the same time** • __Multi-user__… • … allows multiple **users at the same time** (must be clear that candidate is not discussing an OS that simply has multiple accounts) • __Distributed__… • …allows multiple computers to work together **on a single task** • __Embedded__… • …has a dedicated/limited function • …is read-only / cannot be changed	6	Mark in pairs Allow real time if not given as previous answer Do not accept "runs on an embedded system" as expansion of embedded OS, this is NE. __Examiner's Comments__ This question was generally well answered with Embedded, Distributed, Multi-User and Multi-Tasking being the most common answers. Some candidates struggled to name and a type of operating system. Centres should advise candidates that OS brand names are not accepted as a type.
	b		• Interrupt checked for at start/end of each fetch-execute cycle • If the interrupt is of a lower/equal priority to the current process then the current process continues • (If interrupt raised) contents of **registers** copied to stack • Flags are set to determine if interrupts are enabled / disabled • Program counter changed to point to __Interrupt Service Routine (ISR)__ // __ISR__ runs • After interrupt complete, previous **register** values restored back from stack • Flag is reset • If higher priority interrupt received during servicing of interrupt… • …this is added to stack and new interrupt dealt with	3	__Examiner's Comments__ Many candidates were able to gain full marks on this question. Unfortunately, some candidates showed a lack of detail in their answers. Some candidates talked about interrupts being run during an FDE cycle or assumed that an interrupt would be run immediately with no reference to priorities. **OCR support** Resources for operating systems and interrupts can be found in this document. https://www.ocr.org.uk/Images/253685-the-function-and-purpose-of-operating-systemsdelivery-guide.pdf

Question			Answer/Indicative content	Marks	Guidance
	c		**Mark Band 3 – High Level (7–9 marks)** The candidate demonstrates a thorough knowledge and understanding of memory management carried out by operating systems. The material is generally accurate and detailed. The candidate is able to apply their knowledge and understanding directly and consistently to the context provided. Evidence/examples will be explicitly relevant to the explanation. The candidate is able to thoroughly assess the importance of memory management to an efficient and secure system. There is a well-developed line of reasoning which is clear and logically structured. The information presented is relevant and substantiated. **Mark Band 2 – Mid Level (4–6 marks)** The candidate demonstrates reasonable knowledge and understanding of memory management carried out by operating systems. The material is generally accurate but at times underdeveloped. The candidate is able to apply their knowledge and understanding directly to the context provided although one or two opportunities are missed. Evidence / examples are for the most part implicitly relevant to the explanation. The candidate makes a reasonable attempt to assess the importance of memory management to an efficient and secure system. There is a line of reasoning presented with some structure. The information presented is in the most part relevant and supported by some evidence. **Mark Band 1 – Low Level (1–3 marks)** The candidate demonstrates a basic knowledge of memory management carried out by operating systems; the material is basic and contains some inaccuracies. The candidate makes a limited attempt to apply acquired	9	*The following shows example content that may form part of a candidate's answer. It is not intended to be an exhaustive resource, nor should a candidate be expected to specifically cover any particular amount of this.* Knowledge (AO1) • Memory management means to ensure that RAM is used efficiently and not wasted • Removes data not needed anymore (garbage collection), frees up space and allocates memory to applications • Paging or segmentation may be used to split up memory • Paging uses fixed size divisions whereas segmentation uses varying size divisions • Paging is where memory is divided physically • Segmentation is where memory is divided logically • Virtual memory may be used when RAM is (almost)full to enable applications to continue to run Application (AO2) • If RAM is unavailable or full, applications cannot be loaded • Data transferred out of RAM into virtual memory to free up space and then transferred back again when needed • Also includes security so that data stored in memory is not vulnerable • Memory management is important for a well-running machine. If not, RAM would rapidly run out and fill up with unneeded data/instructions and so no new applications could run • Paging causes internal fragmentation whereas segmentation causes external fragmentation • A page table is used to map page location which is slower than a segmentation table • It is easier for the OS to manage page locations as they can be stored non-contiguously. Segments can be non-contiguous but work better contiguously

Question			Answer/Indicative content	Marks	Guidance
			knowledge and understanding to the context provided. The candidate provides nothing more than unsupported assertions. Any discussion of the importance of memory management will be vague or lacking detail. The information is basic and communicated in an unstructured way. The information is supported by limited evidence and the relationship to the evidence may not be clear. 0 mark No attempt to answer the question or response is not worthy of credit		Evaluation (AO3) • RAM is much more expensive than secondary storage (per unit/GB) so virtual memory is useful • rather than having to buy more RAM • Over use of virtual memory causes slow down and even disk thrashing if pages have to be swapped back and forth too often • Paging can be more effective because any free memory space can be used to swap data in and out whereas with segments, lots of space will sit unused until a segment the right size is available • Segmentation errors can cause memory leakage which would cause the system to crash • Security issues – applications can only access memory allocated to them so (for example) a malicious application cannot access the memory allocated to a banking app. Also when applications are closed, data is removed before being reallocated so that applications cannot see historic data <u>Examiner's Comments</u> Many candidates were able to show an understanding of pages being a fixed size and segments being variable size, but few were able to relate virtual memory to the use of pages and segments and few had an understanding of how they are used. Responses to why it is important tended to be vague. There were a few candidates who talked about compression which was not relevant to the question.
			Total	19	

Question			Answer/Indicative content	Marks	Guidance
5			• A piece of software which allows **hardware/device to communicate**… • …with the **operating system**	AO1.2 (2)	<u>Examiner's Comments</u> This question was generally answered well. Some candidates were not given the second mark as they did not make reference to the operating system.
			Total	2	
6		i	• For printer queue • All documents/users have equal priority • Whichever document is received first is printed first • First in First Out / Last in Last Out	2	<u>Examiner's Comments</u> This question was generally well answered with many candidates gaining at least 1 mark. The question asked why the OS used 'first come first served' and many candidates were not given marks for stating why the user would want the OS to use it. Some candidates talked about other scheduling algorithms even though this was not relevant to the question.
		ii	• To enable <u>multitasking</u> to take place • To switch between active processes and those running in the background • To limit each process to a certain amount of time//allow processes an equal share of processor time…. • …to ensure the OS cycles through all processes // the process then goes to the back of the end of the queue • …so that users can receive an immediate response • …to handle an interrupt immediately	3	BP4, 5 & 6 are dependent on BP3 only
		iii	• Shortest job first / shortest remaining time • Process which has the shortest <u>time</u> (remaining) is completed first • Multilevel feedback queues • Uses <u>queues</u> with different priorities • Jobs can be moved between <u>queues</u>	2	One mark for name, one mark for description.
			Total	7	

Question			Answer/Indicative content	Marks	Guidance
7	a	i	1 mark for any of the following points, e.g: • GUI will need to remove open file manager windows • OS will need to remove open files/release locks • Inform the CPU to cease transferring files	1 (AO2.1) (1)	Allow other suitable alternative answers. <u>**Examiner's Comments**</u> Very few candidates achieved a mark here. Many mentioned that the CPU would be busy but were not specific enough to gain the mark. In questions of this type, examples need to be specific to the question.
		ii	1 mark per bullet up to a maximum of 2 marks, e.g: • Stacks use LIFO/FILO storage • As processes are halted by an ISR they are pushed on the stack • When they are returned, they are popped from the top of the stack • So they are returned to in correct order	2 (AO1.2) (2)	<u>**Examiner's Comments**</u> Few candidates gained both marks for this question. The concepts of pushing and popping items was mentioned by many candidates although this was often too vague and didn't link to the question. Candidates needed to use the correct terms for accessing items from a stack to gain the marks in this question.
	b	i	1 mark for any of the following bullet points: • Round Robin • Multi-level feedback queues • Shortest job first • Shortest time remaining	1 (AO1.1) (1)	Do not Allow: • First come First Served <u>**Examiner's Comments**</u> Round Robin was the most popular answer by far. Most candidates achieved the mark although some used the example in the question. Candidates should be reminded to read the questions carefully here and to not repeat answers already provided.
		ii	1 mark per bullet up to a maximum of 2 marks, e.g: • Jobs dispatched on a FIFO basis • Each job/packet is equal priority • Each job/packet switch has the same processing time	2 (AO2.1) (2)	Accept processed instead of dispatched (Bp1) Do not accept task/data instead of jobs (BP1 to 3) <u>**Examiner's Comments**</u> Few candidates achieved both available marks and most missed the concept of jobs/packets having equal priority.

Question			Answer/Indicative content	Marks	Guidance
	c		1 mark per bullet up to a maximum of 2 marks, e.g: • Paging uses physical addressing…. • ….Segmentation uses logical addressing • Paging uses fixed size memory blocks…. • …. Segmentation uses variable length memory blocks	2 (AO1.2) (2)	Answer must cover paging and segmentation for 2 marks. Do not accept data instead of memory Examiner's Comments To get both marks, candidates need to be clear that paging and segmentation are about allocating memory. Many candidates were not clear on this and discussed managing data or packets. Candidates should be careful to use the correct terminology for questions of this type in particular.
			Total	8	
8			1 mark per bullet up to a maximum of 3 marks, e.g.: • Peripheral management • Handle interrupts • File management • Provides a user interface • Provides platform to install and run software. • Provides utilities for system maintenance. • Allows multi-tasking • Provides security	2 AO1.1 (2)	Do not accept memory management or processor scheduling.
			Total	2	
9			– FCFS means jobs are completed in the order they arrive – ineffective in catching viruses / the virus may run first – …the virus checker may never run / take a long time to start running – the virus checker may be continuously running… – …this will temporarily stall the system / all other processes have to wait. (1 mark per -, max 2)	2 AO2.2	Examiner's Comments Those candidates who demonstrated a clear understanding of 'First Come First Served' scheduling scored well on this question. Some candidates incorrectly referred to priorities and interrupts in their responses which gained no credit.
			Total	2	

Question			Answer/Indicative content	Marks	Guidance
10			- Needs to be able to respond instantly to changes… - …such as someone stepping in front of car (or other sensible example) - RTOS offers a guaranteed response time. - A non RTOS might be busy dealing with other tasks and not respond until it is too late. (1 Mark per -, Max 3)	3 (AO1.2)	<u>Examiner's Comments</u> There were three marks available for this question. Candidates should be reminded that they need to specify a clear and specific point for each mark awarded. Many candidates did not extend their responses to three points. In addition, there were many definitions of the form 'the car needs to be able to respond in real time' which gained no credit.
			Total	**3**	
11		i	Paging…(1) …Memory is divided into fixed / physical units (1) Segmentation… (1) …Memory is divided logically / variable size according to its contents. (1)	4 (AO1.1)	Accept same size units for MP1 **Examiner's Comments** Candidates who correctly cited paging and segmentation as the methods of dividing memory, invariably went on to achieve full marks.
		ii	Multitasking allows the user to run more than one program <u>at the same time.</u> (1) E.g. running CAD software whilst checking emails. (1)	2 (AO1.1 – 1 mark AO1.2 – 1 mark)	Accept any reasonable work related answer **Examiner's Comments** Most candidates achieved both marks on this question. Those who did not, either explained multi-tasking **or** gave appropriate examples. The question asked for both.
			Total	**6**	